Nermine Moumen

Final-year Engineering Student in Industrial Chemistry

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EDUCATION

National Institute of Applied Sciences and Technology (INSAT) *2022–Present*

Industrial Chemical Engineering student, specialized in Process and Environment, with a strong foundation in water treatment, analytical chemistry and electrochemical processes. Oriented towards laboratory-scale experiments, process monitoring and data interpretation. Seeking a 6-month master thesis internship starting February 2027.

PROFESSIONAL EXPERIENCE

SEGULA Technologies – Electrochemical Battery Modeling Internship *06/2026–08/2026*

- Modeled a **Li-ion cell** (NMC811/Graphite-SiOx) using the Single Particle Model (SPM) implemented in MATLAB.
- Implemented the electrodes' **open-circuit potential (OCP) curves** and solved the PDEs with **FVM**.
- Optimized the mesh and diffusion coefficient values, reducing the RMSE error **from 202.13 mV to 40.5 mV**.
- Extracted and processed **experimental discharge data** from a database to validate MATLAB and COMSOL results.

Control Union – Quality Control Internship *08/2025–09/2025*

- Performed **laboratory analysis** of lubricants (viscosity, heavy metals by **ICP-OES**) following standardized procedures.
- Prepared and interpreted **analytical reports** to verify compliance with ASTM D5185 and **ISO 17025** standards.

ETAP – Water Treatment and Petrochemistry Internship *07/2024–08/2024*

- Performed physico-chemical and ionic analyses of water samples (pH, salinity, chlorides via ion chromatography).
- Studied water treatment methods **(coagulation-flocculation, ion exchange)** for contaminant and **corrosion** control.
- Performed organic geochemistry techniques **(GC, GC-MS, Rock-Eval pyrolysis)** for hydrocarbon **characterization**.

ADDITIONAL EXPERIENCE

Cascade – Online Chemistry Teacher (Part-time) *2024–Present*

- Provide individualized online chemistry tutoring for French Terminale students, covering exam-level concepts.

ACADEMIC PROJECTS

Electrochemical Treatment of an Industrial Effluent by Electroflocculation

- Carried out **electrochemical measurements** on aluminum **electrodes**: current-potential curves and **pulse polarography**.
- Calculated the **faradaic yield** with **Faraday's law**, linking electrode dissolution to the charge passed.
- Removal of Cd^{2+} and Zn^{2+} from a simulated effluent: **85% and 57%** in 20 min.

Effluent Treatment by Coagulation-Flocculation

- Optimized the coagulation pH (aluminium sulfate / alginic acid) on a **real effluent**, testing 5 pH values; maximum settled floc volume at **pH 8**.
- Interpreted the efficiency through aluminium speciation using the **Pourbaix diagram**.

Energy Recovery from Organic Waste by Anaerobic Digestion (ASBR)

- Experimental follow-up of **5 treatment configurations** (primary, secondary, mixed sludge and co-digestion with fruit and vegetable waste) in **ASBR digesters** (HRT = 20 d).
- Compared performance through COD, VSS, volumetric load, biogas production and CH_4 content, reaching up to **0.50 L/d of biogas at 64.52% CH_4**.

TECHNICAL SKILLS

- **Lab Techniques:** Chromatography (HPLC, GC, TLC), Spectrometry (MS, ICP-OES), Spectroscopy (NMR, FTIR, UV-Vis), Diffraction (XRD), Microscopy (SEM, TEM), Electrochemistry (Cyclic voltammetry, Polarography).
- **Simulation:** Aspen Plus, Aspen HYSYS, COMSOL Multiphysics • **Programming:** *MATLAB, Mathematica, Python*
- **Data & Office:** Microsoft Office Suite, OpenLCA • **Standards:** ISO 9001, ISO 14001, ISO 45001, ISO 17025, ISO 14040

EXTRACURRICULAR ACTIVITIES

- **IEEE INSAT** – Chair of EMBS; Organizing Chair of TRSYP 3.0; Social Activities Chair of MENASYP.
- **THEATRO INSAT** – Member of the dance team.

LANGUAGES AND CERTIFICATES

Languages

English (Fluent), French (Fluent), Arabic (Native).

Certificates

Aspen HYSYS – MATLAB